Quiz-2

ACOL 216: Introduction to Computer Architecture

IIT Delhi - Abu Dhabi · Semester II, 2025-26

(Viva-based Quiz)

Total Marks: 10

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1. Consider a CPU with 256 instructions in the instruction sets, and each operation requires 10 cycles to finish. If 100 control signals are needed to be generated control units and if signal address field format is used, What is the minimum size of control word in the Horizontal and vertical microprogrammed control unit?
   Answer: 112, 19

2. Consider a hypothetical control unit, which supports 1024 bytes of control word memory. If the control word is 30 bits long and hardware contains 4 flags and supports 16 branch conditions, what is the maximum number of control signals that can be generated **at a time** using horizontal micro-programming and vertical micro-programming respectively?
   Answer: 14, 1

3. A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, what is the maximum value of the immediate operand?
   Answer: 16383

4. Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, what is the amount of memory (in bytes) consumed by the program text?
   Answer: 500

5. A processor has 16 integer registers (R0, R1, … , R15) and 64 floating point registers (F0, F1, … , F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type 4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F). What is the maximum value of N?
   Answer: 32

6.

In a look-ahead carry generator, the carry generate function G_i and the carry propagate function P_i for inputs A_i and B_i are given by:

$$P_i = A_i \oplus B_i and\ G_i = A_i B_i$$

The expression for the sum bit S_i and the carry bit C_{i+1} of the look-ahead carry adder are given by:

$$S_i = P_o \oplus C_i \text{ and } C_{i+1} = G_i + P_i C_i, \text{ where } C_\theta \text{ is the input carry}$$

Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gets can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S3, S2, S1, S0 and C4 as its outputs are respectively:

Answer: 10, 4

7. The carry and sum of a full adder takes 20 ns and 30 ns respectively. Then calculate the maximum number of additions performed in a second, if three full adders are cascaded __________ x 10^6 (Round of to two decimal points)

   Answer : 14.29

8. A 4-bit carry look ahead adder, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming that all the inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, what is the overall propagation delay of the adder? Assume that the carry network has been implemented using two-level AND-OR logic.
   Answer : 6 time units

9. Consider a hypothetical control unit that supports 5 groups of mutually exclusive control signals. Also assume that group 1 & group 2 are using horizontal micro-programming whereas group-3, 4 and 5 are using vertical micro-programming. What is the total number of bits used for control words?

Groups	G1	G2	G3	G4	G5
Control Signals	43	29	76	130	100

   Answer: 94

10. A hypothetical cpu supports 300 instructions. Each instruction takes 5 cycles to accomplish the execution. The control unit is designed using vertical programming which has 130 control signals, 64 flags and 12 branch conditions. X and Y represent the number of bits required for control address register (CAR) and control data register (CDR) respectively. What are the values of X and Y?

    Answer: X=11, Y=29

11. Consider a CPU where 150 instructions take 8 clock cycles each to complete the execution. A horizontal microprogrammed control unit has to generate 125 control signals. What is the minimum size of the control word?

Answer: 136

12. Consider a microprogrammed control unit having support of 256 instructions, each of which on average takes 16 micro operations. The system has support of 16 flag conditions and 52 control signals. If vertical microprogramming control is used in the system, then what is the total length of the control word?

    Answer: 22

13. Consider a CPU where all instructions require 10 clock cycles to complete execution. There are 258 instructions in the instruction set. It is found that 129 control signals are needed to be generated by the control unit. While designing the vertical micro-programmed control unit, single address field format is used for branch control logic. What is the size of the control memory (in bytes)?

    Answer: 6450